

TUNIS CARTHAGE, Tunisia

WMO: 607150

Lat: 36.8509N

Lon: 10.2145E

Elev: 22

StdP: 14.68

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 41.6	(c) 44.2	(d) 32.3	(e) 26.8	(f) 53.2	(g) 35.2	(h) 30.1	(i) 51.7	(j) 29.6	(k) 57.9	(l) 26.3	(m) 57.0	(n) 7.6	(o) 200	(p) 0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 19.3	(c) 100.3	(d) 72.2	(e) 96.7	(f) 72.3	(g) 93.4	(h) 72.0	(i) 78.2	(j) 88.5	(k) 77.0	(l) 87.3	(m) 75.8	(n) 86.1	(o) 10.5	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	133.4	82.6	73.8	126.1	81.9	72.9	122.3	81.5	41.6	89.1	40.3	87.3	39.2	86.1	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.9	(b) 22.0	(c) 19.4	(e) 37.5	(f) 108.7	(g) 2.3	(h) 3.6	(i) 35.8	(j) 111.2	(k) 34.5	(l) 113.3	(m) 33.2	(n) 115.3	(o) 31.5	(p) 117.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.9	54.3	54.5	58.5	63.2	70.0	77.5	82.9	83.6	78.0	71.7	62.9	56.5		
	DBStd	11.36	3.83	4.32	4.65	4.16	4.69	5.07	3.94	4.01	4.67	4.80	4.70	4.04		
	HDD50	24	9	10	2	0	0	0	0	0	0	0	0	3		
	HDD65	1284	331	294	208	81	7	0	0	0	0	4	94	265		
	CDD50	6550	143	136	265	397	619	825	1020	1042	840	673	386	204		
	CDD65	2335	0	1	6	28	161	375	555	577	390	212	29	1		
	CDH74	22623	0	7	57	183	1109	3572	6427	6738	3271	1152	100	7		
(13)	CDH80	10092	0	0	11	35	348	1554	3251	3383	1204	297	9	0		
(14)	Wind	WSAvg	8.8	8.7	9.2	9.5	9.9	9.8	9.2	8.2	8.2	7.5	8.0	8.1		
(15) Precipitation	PrecAvg	18.5	2.3	2.2	1.9	1.5	0.9	0.4	0.1	0.4	1.5	2.4	2.1	2.6		
	PrecMax	30.1	6.1	7.5	7.0	5.7	3.2	2.3	1.3	2.9	6.1	7.1	9.0	8.4		
	PrecMin	10.7	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.3		
	PrecStd	4.5	1.4	1.6	1.2	1.1	0.7	0.5	0.2	0.6	1.2	1.7	1.7	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.4	73.4	80.3	84.4	94.4	103.2	105.5	105.6	98.4	91.4	80.8	73.5		
		MCWB	57.1	59.0	61.9	63.6	68.1	71.6	72.5	72.6	72.6	69.7	62.5	58.6		
	2%	DB	66.1	68.2	73.8	78.9	87.6	95.7	99.5	100.2	93.2	86.3	76.9	68.4		
		MCWB	55.9	56.9	59.7	62.3	66.4	69.8	72.3	73.1	73.0	69.7	63.4	58.0		
	5%	DB	63.0	64.8	70.2	75.1	83.6	91.7	95.8	96.6	89.6	82.8	73.6	66.0		
		MCWB	55.0	55.3	58.2	61.0	65.4	69.6	72.1	73.3	73.0	69.6	62.1	57.3		
	10%	DB	61.1	62.4	67.6	71.8	80.3	88.1	93.0	93.2	86.2	80.3	71.3	63.9		
		MCWB	54.0	54.5	57.5	60.0	64.6	69.1	71.8	73.6	72.1	68.9	61.8	56.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.4	60.3	63.9	66.3	70.9	76.4	79.2	80.2	78.9	75.5	69.0	62.9		
		MCDB	65.8	69.0	75.3	77.8	84.8	88.3	90.5	90.5	88.0	82.7	75.0	67.6		
	2%	WB	57.6	58.1	61.3	64.1	69.0	73.9	77.3	78.6	77.0	73.5	66.6	60.1		
		MCDB	63.2	65.7	70.3	74.7	81.3	86.0	88.8	88.9	85.8	80.1	72.7	65.6		
	5%	WB	56.3	56.6	59.7	62.6	67.4	72.3	76.0	77.4	75.6	72.0	64.6	58.5		
		MCDB	61.6	63.1	67.8	71.8	78.6	84.3	87.4	87.7	84.0	79.0	70.6	64.1		
	10%	WB	54.9	55.2	58.1	61.2	66.0	71.0	74.7	76.3	74.3	70.3	63.0	57.2		
		MCDB	60.0	61.3	65.4	69.3	76.8	83.0	86.0	86.7	83.0	77.2	68.9	62.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.4	16.0	16.8	18.2	19.5	19.8	19.3	16.2	15.4	14.4	13.4		
		MCDBR	16.9	19.2	21.8	23.2	24.4	25.1	25.1	24.8	20.3	19.0	18.2	17.1		
		MCWBR	9.7	10.8	11.0	10.5	9.3	8.4	8.7	8.4	8.2	7.8	9.1	9.6		
	5% WB	MCDBR	14.8	17.2	19.5	19.4	20.5	21.5	20.9	20.6	18.3	16.6	15.3	14.9		
		MCWBR	9.5	10.4	10.7	9.8	8.9	8.4	8.5	8.2	8.1	7.8	8.9	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.349	0.368	0.419	0.462	0.480	0.448	0.471	0.462	0.459	0.428	0.384	0.355		
	taud		2.416	2.348	2.191	2.094	2.079	2.208	2.150	2.178	2.195	2.241	2.338	2.402		
	Ebn at Noon		264	272	267	261	258	266	259	258	252	247	247	252		
	Edh at Noon		28	34	43	50	52	45	48	46	43	37	30	27		
(44) All-Sky Solar Radiation	RadAvg		740	990	1374	1738	2062	2323	2342	2054	1555	1183	820	662		
	RadStd		56	68	101	115	111	84	56	87	69	69	46	55		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page